

Abstract

Title: Examining Health Equity Bias in Synthetic Data Generation Techniques

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Background

AI is transforming cancer care by improving diagnosis, treatment, and prevention, despite data privacy challenges. Patient data, crucial for AI, is often siloed due to privacy laws like HIPAA. Synthetic Data Generation (SDG) using generative AI can help, but ensuring privacy and avoiding bias is difficult. My research aims to evaluate state-of-the-art SDG technologies for how they prevent or introduce bias for different subgroups. This involves using advanced machine learning to perform a statistical analysis of healthcare data generated using these techniques.

Hypothesis/Research Question

My research focuses on studying the bias mitigation qualities of recent SDG technologies. In particular, I look at how disparities in the quality of synthetic data between overrepresented and underrepresented subgroups are mitigated or exacerbated by these technologies.

Methods/Proposed Approach

I use a logistic regression and a random forest classifier to predict various target features, such as survey question answers, age, and fitbit traits, that lend well to contextual knowledge of the data. These accuracies will be measured across subgroups. Then, using synthetic data generated from the real data by differentially private SDG technologies, MST [1], PrivBayes [2], Private-GSD [3], the same quality metrics will be taken for the different subgroups in the generated data.

The framework built for this analysis lends itself to work on *any* tabular dataset. I show results in this abstract based on demographic data from a census dataset [4]. Experiments are averaged over 10 trials, and training datasets have 2,000 records. More thorough investigation is currently being taken on *All of Us* data, using a cohort of 559 patients with Acute myeloid leukemia (AML) disease, with the target variable set to predicting income, gender, and age.

Results

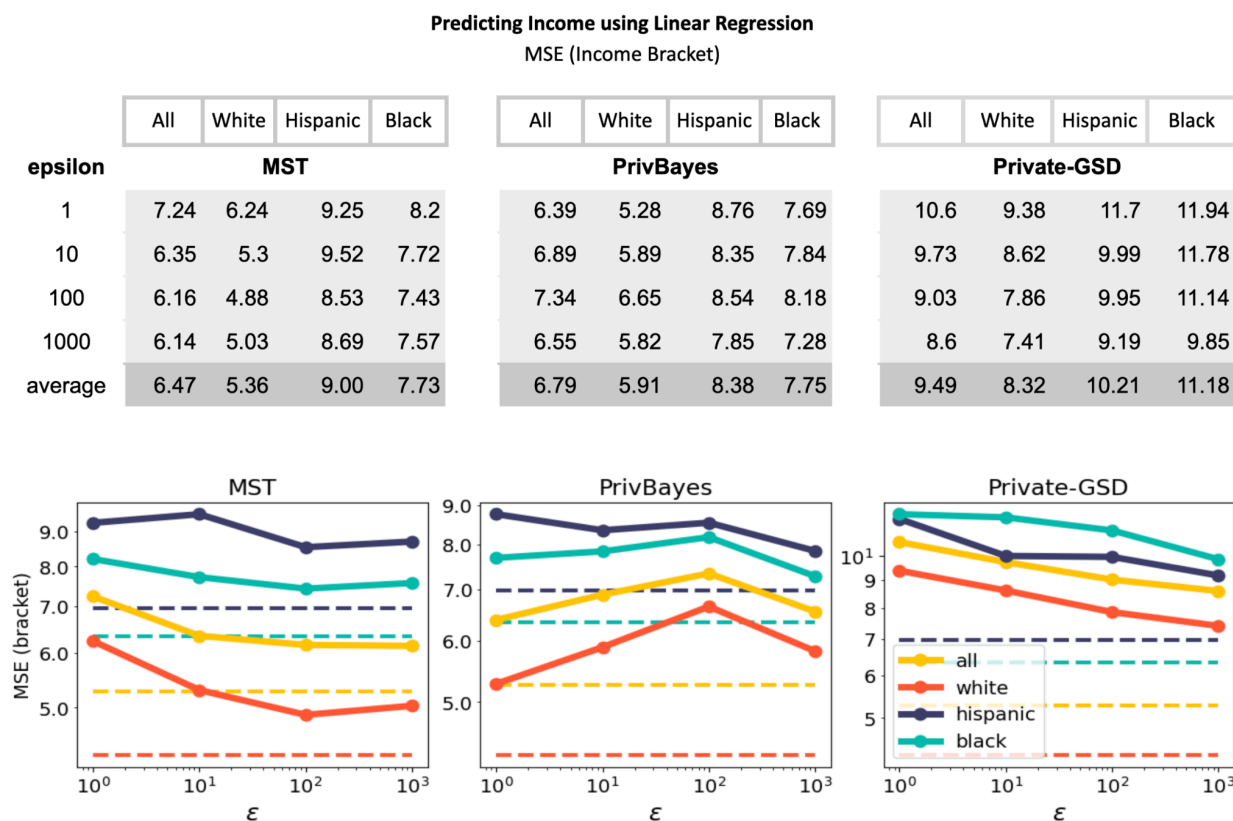


figure 1 mean squared error (MSE) results for predicting income brackets for different subgroups when training a linear regression (LR) model on synthetic data generated by MST, PrivBayes, and Private-GSD, for various privacy levels ϵ . The dashed lines depict the MSE when training on the real data. As hypothesized, predictions made on the smaller subgroups (black, hispanic) yield greater errors generally.

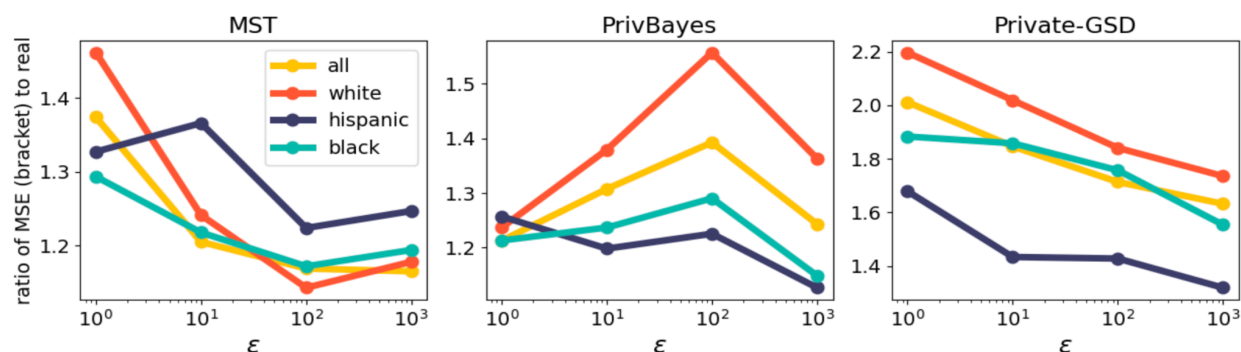


figure 2 proportion between MSE for predicting income brackets when training a LR on synthetic data generated versus when training on the real data. Counter to the hypothesis, the disparity actually *increases* between the white subgroup and the others between discrepancies to the real data's accuracy yield. MST and Private-GSD give fairly consistent disparities across subgroups, across all privacy levels.

Predicting Gender using Random Forest												
Accuracy (%)												
All				All				All				
White				White				White				
Hispanic				Hispanic				Hispanic				
Black				Black				Black				
epsilon	MST				PrivBayes				Private-GSD			
1	60.42	60.78	62.08	61.54	50.94	50.42	52.66	52.72	52.61	52.55	52.18	52.2
10	66.97	66.97	68.47	67.98	65.91	66.72	67.67	66.34	53.88	54.99	54.47	53.92
100	66.92	67.05	68.13	68.72	68.84	69.49	70.26	68.56	54.67	54.99	54.02	54.6
1000	66.1	67.21	68.65	68.72	67.94	68.42	69.31	68.36	53.73	53.39	53.96	53.22
average	65.10	65.50	66.83	66.74	63.41	63.76	64.98	64.00	53.72	53.98	53.66	53.49

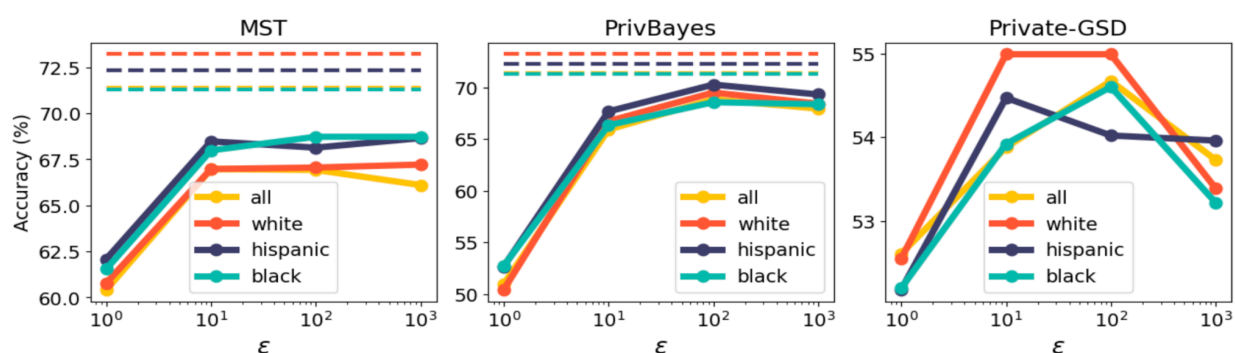


figure 3 accuracy results for predicting gender for different subgroups when training a random forest classifier (RFC) on synthetic data generated by MST, PrivBayes, and Private-GSD, for various privacy levels ϵ . The dashed lines depict the accuracy when training on the real data. Predictions made on data generated by PrivBayes are the most consistent between subgroups.

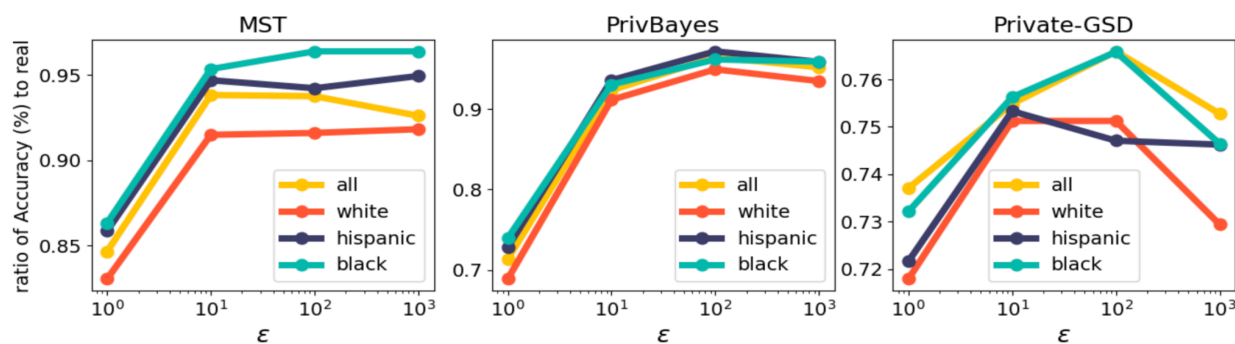


figure 4 proportion between accuracies for predicting gender when training an RFC on synthetic data generated versus when training on the real data. In all three SDGs, the disparities between subgroup accuracies become more pronounced as privacy is weakened (ϵ increased), but also performs more like the real data.

Conclusions

These results show the *proportion* between the accuracy and error of the subgroups in the real data and those of the synthetic data produced by MST, PrivBayes, and Private-GSD, for various values of epsilon. The results indicate that the disparity between subgroups actually decreases in data produced by MST, in general. PrivBayes yields data that generally maintains the disparities found in the real data. However, Private-GSD in fact exacerbates disparities, as can be seen in the increasing proportions of error for the smaller subgroups (hispanic, and then black).

Interestingly, while disparities in gender prediction from the real data are maintained in the SDGs, the SDGs more greatly fluctuate disparities for the numerical feature predictions. So something inherent to each SDG's mechanism mitigates bias differently for different features, which is worth exploring in greater depth.

Questions for Discussion:

1. What about the MST algorithm mitigates the disparity in data quality for underrepresented groups?
2. Why do some SDGs mitigate bias better for only certain types of ML tasks, and not others?
3. What other ML tasks might better measure the respective qualities of synthetic data for subgroups, and for predicting which target variables?
4. How would SDG techniques that are designed to mitigate bias, like DECAF [5], perform under this framework?

References

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